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A **cockroach** can hold its breath for 45 minutes, and can even survive being submerged under water for half an hour. They hold their breath often to help regulate loss of water.

called **alary muscles** (13 pairs). One pair of these muscles is found in each segment on either side of the heart. In cockroach, there is an accessory **pulsatile vesicle** at the base of each antenna which also pumps blood.

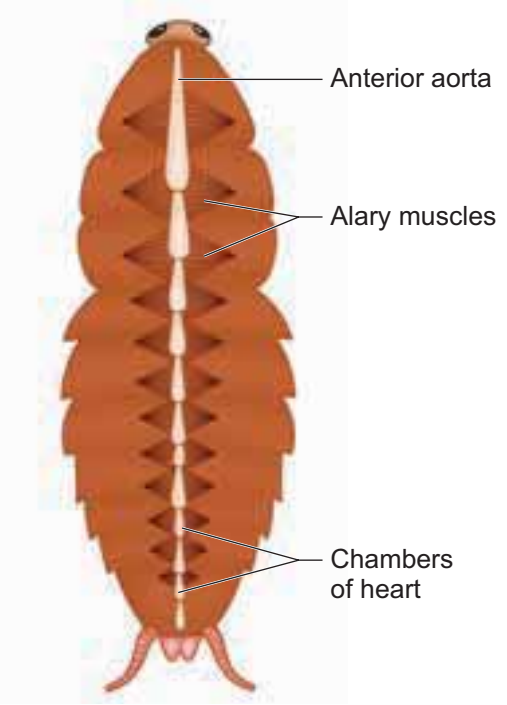


Figure 4.12 *Periplaneta americana*: Circulatory system

Nervous system

The nervous system of cockroach consists of a **nerve ring** and a **ganglionated double ventral nerve cord**, **sub-oesophageal ganglion**, **circum–oesophageal connectives** and **double ventral nerve cord (Figure 4.13)**. The nerve ring is present around the oesophagus in the head capsule and is formed by the supra-oesophagial ganglion called the **'brain'**, The brain is mainly a sensory and an endocrine centre and lies above the oesophagus. Sub-oesophageal ganglion is the motor centre that controls the movements of the mouth parts, legs and wings. It lies below the oesophagus and formed by the fusion of the paired gangalia of mandibular, maxillary and labial segments of the head. A pair of circum–oesophageal connectives is present around the oesophagus, connecting the supra-oesophageal ganglia with the sub-oesophageal ganglion. The double ventral nerve cord is solid, ganglionated and arises from the sub-oesophageal ganglion and extends up to the 7th abdominal segment. Three thoracic ganglia are present, one in each thoracic segment and six abdominal ganglia in the abdomen.

In cockroach, the sense organs are antennae, compound eyes, labrum, maxillary palps, labial palps and anal cerci. The receptor for touch (thigmo receptors) is located in the antenna, maxillary palps and cerci. The receptor for smell (olfactory receptors) is found on the antennae. The receptor for taste (gustatory receptors) is found on the palps of maxilla and labium. Thermoreceptors are found on the first four tarsal segments on the legs. The receptor chordotonal is found on the anal cerci which respond to air or earth borne vibrations. The photoreceptors of the cockroach consists of a pair of compound eyes at the dorsal surface of the head. Each eye is formed of about